

```
In [2]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

Mall Customer Dataset

```
In [3]: df = pd.read_csv("Mall_Customers.csv")
```

Display First 5 Rows

```
In [4]: df.head()
```

```
Out[4]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Understand Dataset

Dataset Shape

```
In [5]: df.shape
```

```
Out[5]: (200, 5)
```

Dataset Information

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                           200 non-null    int64
1   Genre                                 200 non-null    object
2   Age                                   200 non-null    int64
3   Annual Income (k$)                   200 non-null    int64
4   Spending Score (1-100)               200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

Missing Values

```
In [7]: df.isnull().sum()
```

```
Out[7]: CustomerID      0
Genre      0
Age      0
Annual Income (k$)      0
Spending Score (1-100)  0
dtype: int64
```

Statistical Summary

```
In [8]: df.describe()
```

```
Out[8]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

Data Preprocessing

Check Duplicates

```
In [9]: df.duplicated().sum()
```

```
Out[9]: 0
```

Select Features

```
In [10]: X = df[['Annual Income (k$)',  
               'Spending Score (1-100)']]
```

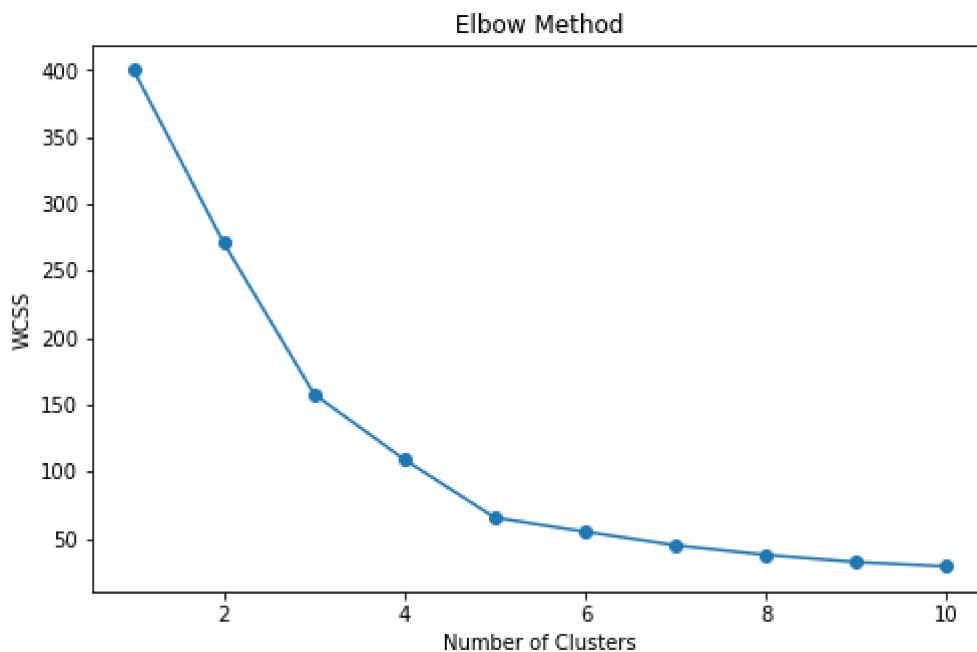
Feature Scaling

```
In [11]: scaler = StandardScaler()  
  
X_scaled = scaler.fit_transform(X)
```

Find Optimal Number of Clusters (Elbow Method)

```
In [12]: wcss=[]  
  
for i in range(1,11):  
    kmeans=KMeans(  
        n_clusters=i,  
        random_state=42  
    )  
  
    kmeans.fit(X_scaled)  
  
    wcss.append(kmeans.inertia_)
```

```
In [13]: plt.figure(figsize=(8,5))  
  
plt.plot(  
    range(1,11),  
    wcss,  
    marker='o'  
)  
  
plt.xlabel("Number of Clusters")  
  
plt.ylabel("WCSS")  
  
plt.title("Elbow Method")  
  
plt.show()
```



Interpretation

Look for the bend/elbow point.

K = 5

Apply K-Means Clustering

```
In [15]: kmeans = KMeans(  
          n_clusters=5,  
          random_state=42  
        )  
  
df['Cluster'] = kmeans.fit_predict(  
    X_scaled  
)
```

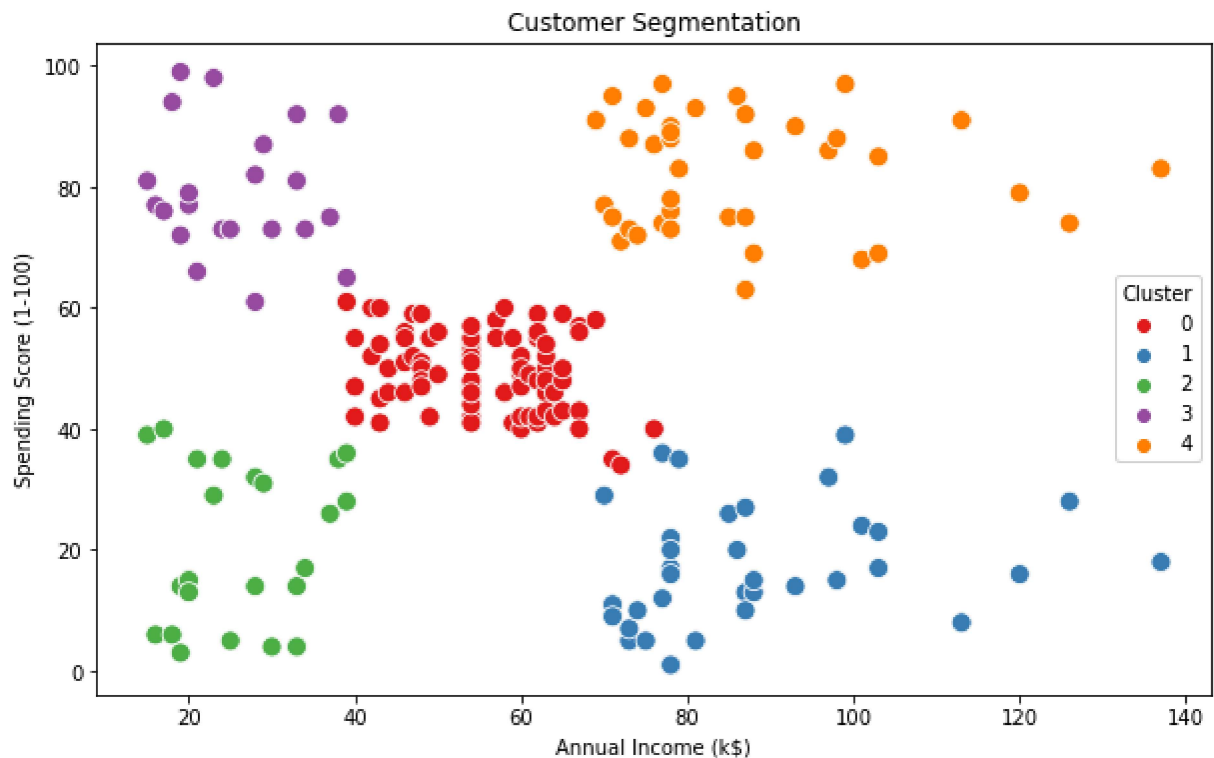
Visualize Clusters

```
In [16]: plt.figure(figsize=(10,6))  
  
sns.scatterplot(  
    x=df['Annual Income (k$)'],  
    y=df['Spending Score (1-100)'],  
    hue=df['Cluster'],  
    palette='Set1',  
    s=100
```

```
)

plt.title(
    "Customer Segmentation"
)

plt.show()
```



Plot Cluster Centers

```
In [17]: plt.figure(figsize=(10,6))

plt.scatter(
    df['Annual Income (k$)'],
    df['Spending Score (1-100)'],
    c=df['Cluster'],
    cmap='rainbow'
)

centers = scaler.inverse_transform(
    kmeans.cluster_centers_
)

plt.scatter(
    centers[:,0],
```

```

centers[:,1],

s=300,

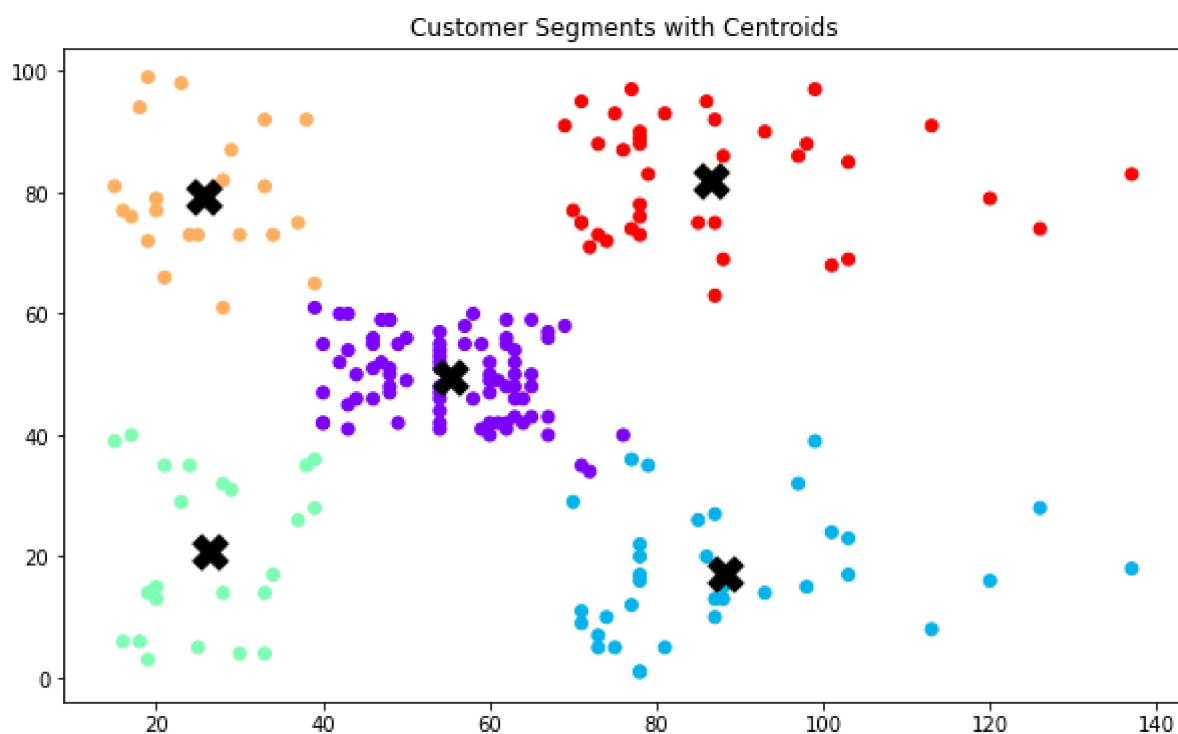
c='black',

marker='X'
)

plt.title(
"Customer Segments with Centroids"
)

plt.show()

```



Analyze Customer Segments

Count customers.

```
In [18]: df['Cluster'].value_counts()
```

```

Out[18]: 0    81
         4    39
         1    35
         2    23
         3    22
         Name: Cluster, dtype: int64

```

View Each Segment

```
In [19]: df.groupby('Cluster').mean()
```

```
Out[19]:
```

	CustomerID	Age	Annual Income (k\$)	Spending Score (1-100)
Cluster				
0	86.320988	42.716049	55.296296	49.518519
1	164.371429	41.114286	88.200000	17.114286
2	23.000000	45.217391	26.304348	20.913043
3	23.090909	25.272727	25.727273	79.363636
4	162.000000	32.692308	86.538462	82.128205

Insights

Cluster 0

Low Income + Low Spending

Budget customers.

Cluster 1

High Income + High Spending

Premium customers.

Cluster 2

High Income + Low Spending

Potential target customers.

Cluster 3

Low Income + High Spending

Impulse buyers.

Cluster 4

Medium customers.

CONCLUSION

The customer segmentation analysis grouped mall customers into different behavioral segments using K-Means clustering.

Five clusters were identified based on annual income and spending score.

High-income high-spending customers were identified as premium customers, while high-income low-spending customers represent potential marketing targets.

The segmentation can help businesses improve marketing strategies and customer targeting.

In []: